BDH-GPU for Searchless Chess

Training a Post-Transformer Architecture on Chess Position Evaluation

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Project: Deep Learning
Proposal Presentation

March 10, 2026

Outline

1. Problem Definition
2. Dataset
3. Evaluation Metric
4. Architecture
5. Engineering Plan
6. Timeline, Risks & Compute
7. Summary

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Searchless Chess: The Problem

Task: Predict chess position quality from a static board state — no tree search allowed.

- **Input:** Board position (FEN notation)
- **Output:** Score $\in [-1, 1]$ (normalized Stockfish centipawn)
- **Formulation:** Supervised regression

No minimax. No alpha-beta. No MCTS. Pure neural intuition.

Prior Art

Model	Params	ELO
DeepMind	270M	2895
ViT-Med*	9.5M	1960
ViT-Small*	2.6M	1817
ResNet-XL*	24.7M	1719

*Grzyb, 2025

Key insight: ViTs achieve 10× better parameter efficiency than ResNets for chess.

My Angle: Why BDH?

Research question:
Can a post-transformer architecture (BDH-GPU) match or beat ViT on searchless chess?

Why BDH suits chess:

1. **Relational state** — piece interactions (pins, forks, batteries) stored as neuron-pair weights
2. **Iterative reasoning** — shared-weight layers applied K times, like "thinking longer"
3. **Sparse activations** — only relevant motifs fire, matching chess pattern recognition

BDH in One Sentence

A recurrent linear-attention model with Hebbian fast-state updates and weight-tied layers.

Novelty

No prior work has applied BDH to chess or any combinatorial game domain.

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Dataset: Lichess-Stockfish-Normalized

Source: HuggingFace (CC BY 4.0)
`huggingface.co/datasets/mateuszgrzyb/lichess-stockfish-normalized`

Property	Value
Total positions	316M
Format	Parquet (10 shards)
Deduplication	FEN-based
Key columns	`fen`, `cp`, `depth`
Original source	Lichess.org

I use: ∼50–100M subset
(compute-constrained to 100h A100)

Input Encoding

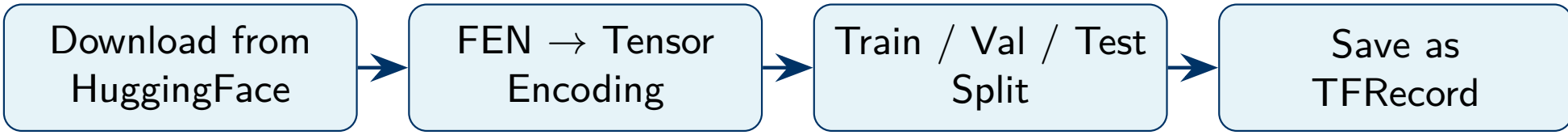
FEN → 8×8×12 tensor

- 64 squares × 12 channels
- Binary piece-type occupancy
- Always-white perspective (board mirrored for black)

Data Split

Train	90%
Validation	5%
Test	5%
Seed	2001

Data Pipeline



Reusable Components

- `fen_to_tensor()` — proven encoding from existing repo
- `mate_to_cp()` — mate → centipawn conversion
- TFRecord streaming for memory efficiency

Engineering Decisions

- **Deterministic** splits (seeded shuffle)
- **Config-driven** — all paths & params in YAML
- **Scripted** download — `datasets` library
- **No raw data** in Git

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Evaluation Strategy

Training Metric:

$$\mathcal{L} = \text{MSE}(\hat{y}, y) = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2$$

Monitoring: MAE for interpretability

Gold Standard: ELO Estimation

1. Solve 1,200 Lichess puzzles (12 tiers, 100 per tier)
2. Accuracy per tier
3. Linear regression fit
4. ELO = rating at 50% accuracy

Why Puzzles Are Valid

Searchless models evaluate only the **immediate position** (depth = 1).

$\Rightarrow$ Puzzle difficulty $\equiv$ pattern recognition complexity

$\Rightarrow$ 50% threshold $\approx$ human-equivalent strength at that rating

Puzzle Dataset

Already available:
`data/puzzles/test_puzzles.feather`

- Ratings: 399 – 3213 ELO
- Proven methodology (Grzyb, 2025)

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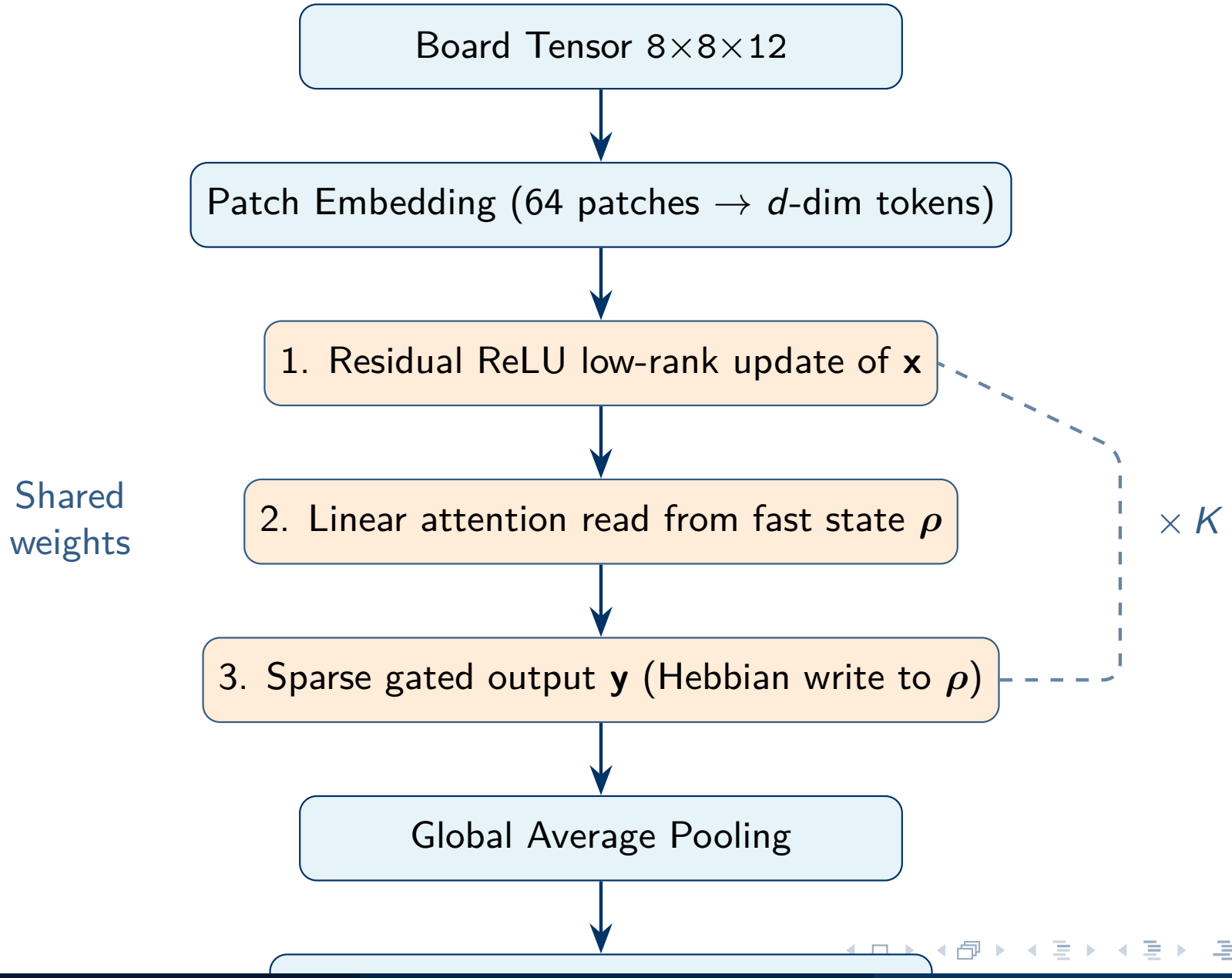
BDH-GPU Architecture



Model Variants & Baselines

BDH-GPU Variants:

Variant	n	K	Params
BDH-Tiny	4096	4	∼1.6M
BDH-Small	8192	8	∼6.3M
BDH-Medium	16384	12	∼12.6M

n = neuron dimension, $d = 256$ (rank),
K = recurrent depth (shared weights)

Key Experiment:
Depth ablation — $K \in \{4, 8, 12\}$
Does more "thinking" help?

Baselines for Comparison:

Model	Params	ELO*
MLP	∼2M	∼900
ViT-Small	2.6M	1817

*Expected / known from prior work

Comparison Strategy

- **Param-matched:** BDH-Tiny vs ViT-Small (∼2.6M)
- **Scale:** BDH-Small/Medium
- **Ablation:** depth K effect

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Repository Structure

Directory	Purpose
`.github/workflows/`	CI/CD pipeline
`configs/`	YAML experim
`src/models/`	BDH, ViT, Res
`src/training/`	Unified training
`src/evaluation/`	Puzzle-based E
`src/data_preparation/`	FEN encoding
`tests/`	pytest test suit
`scripts/`	Entry points
`notebooks/`	EDA & analysi

New Files (my contribution)

- `src/models/bdh.py` — BDH-GPU implementation
- `src/training/trainer.py` — unified config-driven training
- `src/evaluation/puzzle_evaluator.p` — ELO estimation
- `tests/test_bdh_model.py` — forward pass & shape tests
- `tests/test_training_smoke.py` — 1-batch smoke test

DevOps Toolchain

Environment: uv

```
uv sync
uv run python scripts/training/train.
  --config configs/train_bdh_small.ya
```

CI/CD (GitHub Actions)

- `ruff check` — linting
- `ruff format -check` — formatting
- `pytest tests/ -v` — all tests
- Triggered on push & PR to `main`

Testing (pytest)

- Data loading & encoding
- BDH forward pass & output shape
- Loss computation sanity check
- 1-batch training smoke test
- Evaluation pipeline test

Bonus Targets

- Docker container
- Streamlit demo
- Inference benchmarking
- W&B Sweeps

Reproducibility & Experiment Tracking

Reproducibility Guarantees:

Requirement	Implementation
Random seeds	YAML → `tf`, `np`, `random`
Fixed splits	Seeded shuffle, saved to JSON
Config-driven	1 YAML per experiment
Best model	`ModelCheckpoint`
Pinned deps	`uv.lock`

Unified Entry Point:
One script, any architecture:

```
uv run python scripts/training/train.
  --config configs/train_bdh_small.ya
```

Weights & Biases Integration:

- All hyperparams logged automatically
- Training curves (loss, MAE)
- ELO estimates per experiment
- Model artifacts saved

Planned Experiments ($\geq$7)

1. MLP baseline
2. ViT-Small baseline
3. BDH-Tiny (K=4)
4. BDH-Small (K=8)
5. BDH-Small depth ablation
6. BDH-Medium (K=12)
7. BDH + adaptive depth

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Compute Budget (100h A100)

Task	Hours	Cumulative	Phase
Data pipeline + encoding	2h	2h	Infrastructure
MLP baseline	2h	4h	Baselines
ViT-Small baseline	5h	9h	Baselines
BDH-Tiny (fast iteration)	5h	14h	BDH Development
BDH-Small (main model)	20h	34h	BDH Development
Depth ablation (K=4,8,12)	15h	49h	Ablations
BDH-Medium (scale-up)	40h	89h	Advanced
Evaluation + misc	5h	94h	Evaluation
Buffer	6h	100h	Safety

BDH-Medium is expendable — if behind schedule, reallocate to more ablations.

Meeting Roadmap

#	Meeting	Deliverable	Engineering Focus
3	Proposal	This presentation	—
4	Repo & CI	Public repo, CI green	`uv`, `ruff`, `pytest`
5	EDA + Baseline	Notebooks, MLP/ViT	Data pipeline tests
6	Reproducible	Config training, W&B	Seeds, checkpoints
7	First BDH	BDH-Tiny/Small results	W&B $\geq$3 experiments
8	Improvements	Tuning, depth ablation	Before/after comparison
9	Error Analysis	Failure by puzzle tier	Overfit/underfit analysis
10	Advanced	BDH-Medium, ablations	W&B $\geq$7 experiments
11	Final Review	Clean repo, README	CI green, all tests pass
12	Presentation	15–20 min final	Full results + conclusions

Risks & Mitigations

High Risk

BDH underperforms ViT

Mitigation: Still valid as architecture comparison study. Strong baselines ensure meaningful contribution regardless.

High Risk

No reference implementation

Mitigation: Start with BDH-Tiny, extensive unit tests, verify gradient flow early. *(Also: bonus for engineering quality.)*

Medium Risk

Compute overrun

Mitigation: Strict hour tracking. BDH-Medium is expendable.

Low Risk

Data pipeline issues

Mitigation: Reuse proven pipeline. HuggingFace `datasets` handles downloading.

Fallback Strategy

If BDH fails to converge → simplify (remove Hebbian updates, reduce to linear attention) → project becomes "Lessons Learned" with strong engineering.

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Summary

1. **Problem:** Searchless chess — predict position quality without search
   Supervised regression: FEN $\rightarrow$ score $\in [-1, 1]$
2. **Dataset:** 316M positions (Lichess-Stockfish-Normalized, HuggingFace)
   Proven to produce 1960 ELO models
3. **Architecture:** BDH-GPU — post-transformer with Hebbian state
   3 scales (1.6M–12.6M params), depth ablation $K \in \{4, 8, 12\}$
4. **Metric:** MSE (training) + ELO via 1200 puzzle benchmark
   Direct comparison to ViT-Small (1817 ELO)
5. **Engineering:** `uv` + CI/CD + `ruff` + `pytest` + W&B + YAML configs
   Config-driven, reproducible, $\geq$7 tracked experiments

Compute: 100h on NVIDIA A100

Thank You!

Questions?

Dataset: `huggingface.co/datasets/mateuszgrzyb/lichess-stockfish`

Reference repo: `github.com/mateuszgrzyb-pl/searchless-chess`

BDH paper: Available upon request

Input: $8 \times 8 \times 12$ board tensor $\xrightarrow{\text{BDH-GPU } (K \text{ passes})}$ Output: position score $\in (-1, 1)$

Backup: BDH-GPU Core Equations

Trainable parameters: $E, D_x, D_y \in \mathbb{R}^{n \times d}$
Total: $\approx (3 + o(1)) \cdot n \cdot d$

Per-layer recurrence (simplified):

1. **Update activations:** $\mathbf{x}_l \leftarrow \mathbf{x}_{l-1} + \mathrm{ReLU}(D_x \cdot \mathbf{v})$
2. **Read from fast state:** $\mathbf{a}_l = E \cdot \rho_{l-1} \cdot \mathbf{x}_l$
3. **Produce sparse output:** $\mathbf{y}_l = \mathrm{SparseGate}(\mathbf{a}_l)$
4. **Hebbian state update:** $\rho_l = \rho_{l-1} + \mathbf{v}_l^* \cdot \mathbf{x}_l^T$ where $\mathbf{v}_l^* = E^T \mathbf{y}_l$

Key property: Same weights at every layer $\rightarrow$ variable depth K is free.

Backup: YAML Config Example

```yaml
experiment:
  name: "bdh-small-k8"
  seed: 2001
  wandb_project: "bdh-searchless-chess"

model:
  architecture: "bdh"
  n_neurons: 8192
  d_rank: 256
  k_depth: 8
  dropout: 0.1

training:
  epochs: 100
  learning_rate: 1e-4
  min_learning_rate: 1e-6
  weight_decay: 1e-4
  batch_size: 512
  scheduler: "cosine"
  mixed_precision: true
```

Backup: ELO Estimation Method

1. **1,200 puzzles** across 12 difficulty tiers (399–3213 ELO)
2. For each tier: accuracy $= \frac{\text{correct puzzles}}{100}$
3. Fit linear regression: Rating $= f(\text{Accuracy})$
4. **ELO estimate** = predicted rating at 50% accuracy

Example from prior work (Grzyb, 2025):

Model	**Tier 4** (1000)	**Tier 6** (1500)	**Tier 8** (2000)
CNN (1.18M)	43%	22%	7%
ViT-Small (2.6M)	82%	68%	34%
ViT-Medium (9.5M)	88%	86%	45%